XIAMEN UNIVERSITY MALAYSIA

ONLINE FINAL EXAMINATION

Course Code:	EEE202		
Course Name:	Engineers and Society		
Question Paper Setter:	Liew Chia Pao		
Academic Session:	2022/04	Question Paper:	A ☑ B ☐
Total No. of Pages:	4 (excluding cover page)	Time Allocated:	2 hours
Additional Materials:	-		
Apparatus Allowed:	-		

INSTRUCTIONS TO CANDIDATES

1. This paper consists of two (2) sections:

Section A:	60 marks	Two (2) questions of 30 marks each
Section B:	40 marks	One (1) question of 40 marks

2. Answer all questions.
3. Your answers must be handwritten.
4. Please follow the requirement of each question and paste on the downloadable answer book provided.
5. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.
6. Section A will be released at 9:30 AM; and Section B will be released before 10:30 AM.

(Student ID: _____ Full Name: _____)

SECTION A

Question 1 (30 marks) – CLO1: C3

Consider the ethical case in Scenario Q1. Use the Guidelines for Code of Professional Conduct to support your answers.

Scenario Q1

Adrian, a fire prevention consultant, was hired by Company X to prepare a private and confidential report on the renovation of a 30-year-old residential apartment in a town with about 200 families. Adrian tested the fire sprinkler system within the occupied residential units. The test revealed that the sprinkler system has defective control valves in all of the residential units, which violates the local fire code. Adrian informed Company X of the test results and the code violation. Adrian was informed in a follow-up meeting with the company that the renovation project will be postponed, which means that the problem with the fire sprinkler system will not be addressed immediately, but rather when funding becomes available in the future. Adrian received payment for his consulting services.

a) Identify the key points of the case.

(6 marks)

b) Identify four (4) subsections or section(s) of the Code of Professional Conduct applicable to the case. Explain why the identified subsections or section(s) are relevant.

(16 marks)

c) Discuss Adrian's ethical obligations under the circumstances.

(8 marks)

Question 2 (30 marks) – CLO1: C3

Consider the ethical case in Scenario Q2. Use the Guidelines for Code of Professional Conduct to support your answers.

Scenario Q2

Benson works for an insurance company, and part of his job entails visiting manufacturing plants to assess compliance with the company's best practises, as well as compliance with codes, standards, and government regulations. Benson begins the day at Manufacturing Z with a meeting with facility management to review prior visits and documentation of safety records. Management of Manufacturing Z brought their safety compliance specialists to the plant to assist with the visit, and to respond quickly to Benson's findings. As Benson prepares to begin plant inspection, he realises he has forgotten to bring the necessary Personal Protective Equipment (PPE) - his safety boots. Manufacturing Z employees offer him a new pair of safety boots; they routinely provide boots to visitors who do not have the necessary PPE. Benson's company has a policy that prohibits accepting gifts worth more than RM100, and because he has recently purchased new safety boots, Benson is aware that the boots Manufacturing Z is offering are worth more than RM100. If he returns to his hotel to get his own boots, he will be unable to complete the inspection, and the visit will have to be rescheduled because the plant owner's safety specialists will be unavailable the following day.

a) Identify the key points of the case.

(10 marks)

b) Identify two (2) subsections or section(s) of the Code of Professional Conduct relevant to the case under study. Explain why the identified subsections or section(s) are relevant.

(10 marks)

c) What should Benson do? What other options does he have if he chooses to wear the boots provided by Manufacturing Z?

(10 marks)

SECTION B

Question 3 (40 marks) – CLO3: C4

a) At a United Nations conference, it was suggested that "the rate of growth of the world's population is the biggest problem that we will face in the millennium".

i. Comment on this statement and discuss some of the likely environmental consequences of human overpopulation.

(9 marks)

ii. Figure Q3 shows the population pyramid for three countries by Pearson Education, Inc. Explain the variances in these countries based on the demographic transition model.

(12 marks)

b) In the United Nations 2030 Agenda for Sustainable Development, climate change was addressed as "one of the greatest challenges of our time" and "its adverse impacts undermine the ability of all countries to achieve sustainable development" became a pressing concern.

i. Explain on the relationship between climate change and sustainable development.

(6 marks)

ii. Explain the possible approaches in engineering practice (solutions) to effectively meet the "Sustainable Development Goal 7: "Ensure access to affordable, reliable, sustainable and modern energy for all" and "Sustainable Development Goal 13: Take urgent action to combat climate change and its impacts".

(13 marks)

Question 3 (Continued)

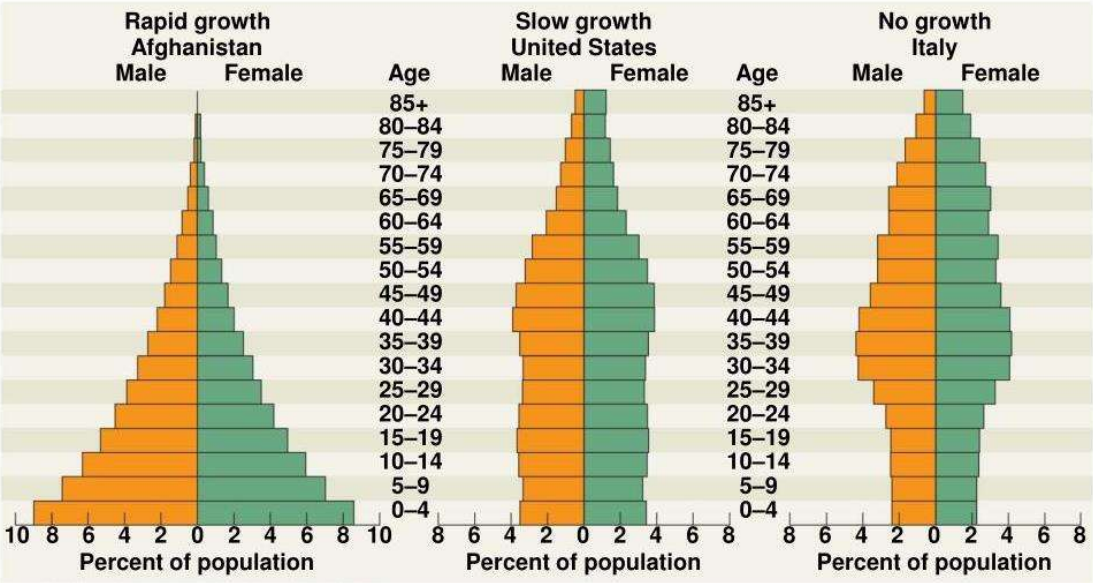


Figure Q3

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